GG-R-1971
March 24, 1976

# FEDERAL SPECIFICATION

# RETRACTOR, RIB, HARKEN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE

1.1 The rib retractor covered by this specification is of the Harken design with two fixed blades and two detachable scapula blades to be used in thoracic surgery.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

Federal Specifications:

| | | |
|---|---|---|
| GG-I-526 | - | Instruments, Dental and Surgical; General Specification. |
| PPP-B-566 | - | Boxes, Folding, Paperboard. |
| PPP-B-585 | - | Boxes, Wood, Wirebound. |
| PPP-B-601 | - | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | - | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | - | Boxes, Shipping, Fiberboard. |

Federal Standards:

| | | |
|---|---|---|
| Fed. Std. No. 66 | - | Steel: Chemical Composition and Hardenability. |
| Fed. Std. No. 123 | - | Marking for Shipment (Civil Agencies). |
| Fed. Std. No. 151 | - | Metals, Test Methods. |

FSC 6515

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

| | | |
|---|---|---|
| MIL-P-116 | - | Preservation, Methods of. |
| MIL-W-6858 | - | Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys, Spot and Seam. |
| MIL-W-8611 | - | Welding, Metal Arc and Gas, Steels and Corrosion Resistant Alloys, Process for |
| MIL-B-10377 | - | Boxes, Wood Cleated, Veneer, Paper Overlaid. |
| MIL-L-10547 | - | Liners, Case and Sheet, Overwrap, Water-Vaporproof, or Waterproof, Flexible. |

Military Standards:

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-109 | - | Inspection Terms and Definitions. |
| MIL-STD-129 | - | Marking for Shipment and Storage. |



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(Copies of Military Specifications, Standards and Publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

Uniform Classification Committee:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

American Iron and Steel Institute (AISI):

Steel Products Manual

(Applications for copies should be addressed to the American Iron and Steel Institute, 1000 16th Street, N.W., Washington, DC 20036.)

American Society for Testing and Materials (ASTM) Standard:

E-18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)



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## 3. REQUIREMENTS

3.1 Materials. Materials shall be as specified herein. When materials other than those specified are used, the supplier shall certify to the procuring activity that the substitute material enables the instrument to meet the requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the product. When a definite material is not specified, a material shall be used which will enable the instrument to meet the requirements of this specification.

3.1.1 Metal parts. All metal parts shall be as specified herein and in accordance with the applicable requirements of GG-I-526. Rack and retractor arms shall be fabricated from American Iron and Steel Institute (AISI) type 400 series stainless steel. The handle and scapula blades shall be fabricated from AISI type 300 series stainless steel.

3.2 General requirements.

3.2.1 Description. The instrument described in this specification is similar or equal to first class rib retractors of the Harken design now manufactured and in general use. It shall have two fixed blades and two detachable scapula blades which shall be easily attached to or removed from the rib retractor arms. The retractor arms shall be designed to fit the contours of the body and shall have a series of equidistant spaced holes (centered to match pins on scapula blade) in each arm, through which the pins of the scapula blade are inserted. The retractor shall be of the self-retaining type with a rack and pinion arrangement which shall operate smoothly. The moveable arm of the retractor shall travel freely along the rack. When the rack and pinion are engaged, at any position from closed to full open, the moveable arm shall be locked in position and immoveable when pressure is applied to the retractor blades. The swivel handle and pinion shall be designed so that they are removable allowing the moveable retractor arm to be removed

3.3 Design. The Harken rib retractor shall be designed to meet all the requirements and tests specified herein and shall conform to the requirements of GG-I-526, where applicable

3.3.1 Mechanical dimensions. The instrument assembly and component parts shall conform to figures 1 to 7, inclusive.



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3.3.2 Rack and pinion locking. The rack and pinion shall be designed so as to be properly aligned and undercut for safe locking. The pinion shall be designed to insure easy and positive engagement. The pinion shall move freely and hold the blade at any position from closed to full open. The moveable retractor arm shall be locked in position and shall not move when pressure is applied to the retractor blade when tested as specified in 4.5.2.

3.3.3 Swivel handle and pinion assembly. The handle and pinion assembly shall be designed so that they are removeable to facilitate disassembly and removal of the sliding retractor arm. The assembly shall have two-degrees of freedom. The handle shall pivot through 180° and provide a continuous rotation of 360° to the pinion upon engagement with the rack. It shall be designed as a replaceable item, and shall be assembled with a suitable attachment method which will allow replacement of either part.

3.3.4 Hardness. The rack and retractor arms shall have a Rockwell hardness of C40 to C45 after being properly stress relieved. The handle and scapula blades shall have a Rockwell hardness of B75 to B80. Hardness shall be tested as specified in 4.5.3.

3.3.5 Corrosion resistance. The instrument shall be corrosion resistant conforming with requirements 3.7.1 and 3.7.3 of GG-I-526 when tested as specified in 4.5.4.1 and 4.5.4.2.

3.4 Finish. The finished instrument shall have a satin finish in accordance with 3.6.2 of GG-I-526, except that the scapula blades may have a scratch brush finish on the top surface and may be rough ground on the back surface.

3.4.1 Passivation. As a final finish operation instrument parts fabricated from AISI type 300 or 400 series materials shall be passivated.

3.5 Identification marking. Identification marking of instruments made in the U.S.A. and of foreign (other than U.S.) manufacture shall be marked "stainless". Requirement 3.8 of GG-I-526 shall also apply. In addition each component part shall be identified with the manufacturer's name.



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3.6 Interchangeability and substitutability. Interchangeable parts having the same manufacturer's part number shall be functionally and physically interchangeable. Component parts need not be interchangeable between manufacturers and may not be interchangeable with different models produced by the same manufacturer.

3.7 Welding. All welds shall be continuous. Welding of steels and corrosion resistant alloys shall be in accordance with MIL-W-8611. Welding shall be durable, sound in every detail and in accordance with latest practices recommended by the American Welding Society. Welding shall be performed only by operators experienced in performing this type of weld, and using the materials required by this specification. Prior to engaging in production welding, the welder shall demonstrate satisfactory workmanship by means of tests specified in 4.5.

3.7.1 Preparation of surfaces. Prior to welding, surfaces shall be free from grease, coatings or other foreign matter which could adversely effect the weld. The edges to be joined shall be filed or machined smooth. Rough, torn, or sheared edges are not suitable.

3.7.2 Joints. All weld joints shall comply with figure requirements respecting type, location and dimensions where specified. Where practicable, all weld joints shall flare into the adjoined metal in gradual smooth curves, and be sealed on all sides. Beads shall be smooth and free of slag. Burn-through and undercut shall not be acceptable. If fluxes are used, the excess shall be removed by a suitable method promptly after welding. There shall be no visible evidence of cracks, voids, porosity or inclusions in the weld, nor evidence of burning or overheating in the base metal. The fillet weld head contour shall be flat or slightly convex. Weld beads shall not be concave. There shall be no notches (incomplete fusing) along the length of the weld.

3.7.3 Weld Test Specimen failure. Failure of the Weld Test Specimen to comply with 4.5.1, 4.5.1.1, 4.5.1.2, and 4.5.1.3 shall constitute disqualification of the welding operator for production welding and rejection of all units represented by the weld test specimen.

3.8 Workmanship. Workmanship shall be first class throughout The instrument shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect life, serviceability or appearance.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing determination by technical means by physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the corrosion-resistant metals may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For weld inspection of retractor arm.

4.2.2 Weld Test Specimen. Prior to producing welded retractor arm assemblies, each welding operator shall prepare one Weld Test Specimen for weld inspection and approval. Only previously approved operators shall produce welded assemblies.



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During production of welded assemblies, each operator shall prepare one Weld Test Specimen for each uniform lot of welded assemblies completed. The Weld Test Specimen shall be inspected and approved prior to continuing production. Non-approval of the Weld Test Specimen shall constitute Inspection Rejection of the lot of complete assemblies for either rework or final rejection. The Weld Test Specimen shall consist of two pieces of production material used to fabricate the retractor arm:

(a) one 5/32" x 27/32" x 8" piece of arm material

(b) one 12" x 8" piece of blade material, formed in accordance with figure 7, Detachable Scapula Blade

The arm and blade pieces shall be welded along one, approximately 3-1/16" edge in the same fashion as the arm is welded to the blade in production. When completed, it shall exactly simulate the end section cut from a production retractor arm unit, and shall have received the same cleaning, welding and finishing procedures as this unit.

4.2.3 Preparation for delivery. The inspection lot size shall be expressed in units of one fully prepared shipping container, containing complete instruments fully prepared for delivery and made from essentially the same materials and components. The sample unit shall be one fully prepared shipping container, containing instruments fully prepared for delivery, with the exception that it need not be sealed. The inspection shall be Inspection level S-2 and the Acceptable Quality Level shall be 4.0 percent defective in accordance with MIL-STD-105.

4.2.4 Sampling for examination. The inspection lot shall be determined in accordance with MIL-STD-105. A random sample of instruments shall be selected from each lot offered to the Government in accordance with MIL-STD-105. The Acceptable Quality Levels (AQL'S) shall be as indicated in table I. The Unit of product shall be one complete rib retractor.

TABLE I. Sampling for examination

| Type defect | Inspection level | AQL (percent defective) |
|---|---|---|
| Major | II | 1.0 |
| Minor | II | 2.5 |



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4.2.5 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-106. Unless otherwise specified the unit of product for tests shall be one complete rib retractor.

TABLE II. Sampling criteria for tests

| Component part | Characteristic | Requirement | Test method | Inspection level | AQL |
|---|---|---|---|---|---|
| Weld test specimen | Weld fracture | 3.7 | 4.5.1.1<br>4.5.1.2 | 4.2.2 | 1/ |
| Retractor assembly | Locking, rack & pinion | 3.3.2 | 4.5.2 | | |
| | Hardness | 3.3.4 | 4.5.3 | | |
| | Corrosion resistance | 3.3.5 | 4.5.4.1<br>4.5.4.2 | S-2 | 2/ |
| Materials | Chemical composition | GG-I-526 | 4.5.5 | --- | --- |

1/ Acceptance criteria shall be as specified in 4.2.2.
2/ Acceptance number shall be zero for all sample sizes.

4.2.6 Raw materials. A sufficient quantity of each lot of each metal used in the construction of the retractor shall be selected for chemical analysis in accordance with GG-I-526.

4.2.7 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The rib retractors shall be examined to determine compliance with all requirements of this specification. Examination shall be conducted as specified in table I and table III. Examination, however, shall not be restricted to defects listed in table III. Any rib retractor in the sample containing one or more defects shall be rejected, and if the number of defective rib retractors in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.



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TABLE III. Classification of defects

| Category | Defect |
|---|---|
| Critical: | None defined. |
| Major: | |
| 101 | Any component missing, cracked or broken, malformed or not functioning properly. |
| 102 | Any component improperly assembled or welded. |
| 103 | Weld bead not smooth and free of slag. |
| 104 | Fixed blade weld burn-through and undercut. |
| 105 | Visible evidence of cracks, voids, porosity or inclusions in the weld. |
| 106 | Visible evidence of burning or overheating of the base metal. |
| 107 | With pinion disengaged, arm does not slide smoothly over full range of travel. |
| 108 | Swivel handle and pinion are not removeable; operation not smooth. |
| 109 | Moveable retractor arm cannot be disassembled from unit. |
| 110 | Rack and pinion do not engage and disengage properly. |
| 111 | With pinion engaged, moveable arm not locked in position. |
| 112 | Rack and pinion not free of sharp edges and burrs. |
| 113 | Blades have sharp edges on bearing face. |
| 114 | Detachable blade fit not as specified. |
| 115 | Material not as specified. |
| Minor: | |
| 201 | Parts not finished as specified. |
| 202 | Mechanical dimensions not within tolerance specified. |
| 203 | Workmanship not as specified. |
| 204 | Fixed blade holes not round. |

4.4 Examination of preparation for delivery. Each of the fully prepared shipping containers, containing instruments, selected as a sample unit from the lot shall be visually examined to determine whether the packaging, packing and marking conform to this specification. The classification of defects, table IV, shall be used to classify the defects found.



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4.4.1 Examination of closure, waterproofing and banding of contents. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

TABLE IV. Examination of defects for preparation for delivery

| Examine | Defects |
|---|---|
| Markings | Omitted, incorrect, illegible, or improper size, location, sequence or method of application. |
| Materials | Any component missing; any component damaged, affecting serviceability. |
| Workmanship | Inadequate application of components such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. |
| | Bulging or distortion of containers. |
| Contents, exterior & interior | Number of intermediate packages is more or less than required; gross/net weight exceed requirements. |

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Visual inspection. The weld test specimens shall be examined for compliance with all requirements of 3.7 and subparagraphs thereto.

4.5.1.1 Destructive testing. Following the visual inspection of 4.5.1 the Weld Test Specimen shall be cut into five similar pieces, by making four transverse cuts on the 8" length. Each piece shall have a fixed blade and arm section measuring approximately 5/32" x 27/32" x 1-1/2" and 1-7/8" x 1-1/2" respectively and shall be a miniature replica of the original test specimen. Both of the end pieces shall be discarded. Two of the three center pieces shall be used for fracture inspections of the weld area; one set of fractures shall be produced by bending the fixed blade away from their fillet welds, and the other set, by bending the fixed blade toward the fillet weld. In each case the moment of bending shall be at the weld areas. If fracture does not occur at a 45° bend of the blade, the sample shall be accepted as passing the test and no further inspection shall be required. Welds which fracture easily or at bending of 45° or less shall be unacceptable.



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4.5.1.2 Weld fracture inspection. The weld fractures shall be visually inspected under a magnification of 10 diameters. The fracture shall be free of excessive porosity, discoloration, inclusions and evidences of cracks, laminations, overlapping or undercutting. The weld shall exhibit evidence of adequate fusion with respect to root and also joint penetration.

4.5.1.3 Microscopic inspection. The third center piece of Weld Test Specimen from 4.4.1.1 shall have one of its saw-cut surfaces prepared by grinding and polishing methods suitable for microscopic inspection of the weld area as seen on cross section. Magnification shall be nominally at 10 diameters. Either Kellers or other suitable etches shall be used to reveal the fusion zone. With reference to suitable depth of weld penetration 50 to 75% of the arm shall be considered adequate. The microscopic inspection procedure need be followed only on the weld test specimen, if failure results on the test identified as Weld Fracture Inspection (4.5.1.2).

4.5.2 Locking and retention test. The rib retractor assembly shall be installed in a fixture that will hold the fixed retractor arm in a fixed position. The detachable scapula blades need not be assembled to the unit. Move the sliding retractor arm moved to an open position approximately 7-1/4" (measured from the inside edge of the fixed arm to the inside edge of the moveable arm). Apply 75 pounds total load to the bearing face of the scapula (fixed) blade. Movement or deflection of the arm shall be cause for rejection. Repeat the test at the 4" and 2" open positions.

4.5.3 Hardness. Rockwell hardness shall be determined in accordance with ASTM test method E18. A minimum of 5 hardness readings shall be made on test specimens representative of the material used in the fabrication of the instrument.

4.5.4 Corrosion tests.

4.5.4.1 Copper sulfate test. The instrument shall be subjected to the copper sulfate test and tested by the following method: Each sample shall be scrubbed with soap and warm water, rinsed in hot water, followed by a dip in ethyl alcohol (95%), or National Formulary (NF) or American Chemical Society (ACS) isopropyl alcohol (isopropanol; 2-propanol), and dried. Instruments, whenever possible, shall be completely immersed in the copper sulfate solution at a temperature of $65^{\circ} + 2^{\circ}$ F. The instrument if too large for complete immersion shall have



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partial immersion or test by drops of this solution. Copper sulfate solution shall be allowed to remain in contact for 6 minutes and shall then be wiped off with a cloth saturated with fresh water. The copper sulfate solution shall be made up as follows:

| | |
|---|---|
| Copper sulfate ($CuSO_4 \cdot 5H_2O$) | 4.0 grams |
| Sulphuric acid ($H_2SO_4$, sp. gr. 1.84) | 10.0 grams |
| Water ($H_2O$) | 90.0 ml. |

There shall be no plating of copper on the instrument. Copper plating at the periphery of the drops of the copper sulfate solution, or at soldered, brazed or welded joints, or dulling of polished surfaces caused by the copper sulfate solution, shall be disregarded.

4.5.4.2 Boil test. The instrument shall be subjected to the boil test and tested by the following method: Before testing, instruments shall be scrubbed with soap and water, rinsed thoroughly in distilled water, and dried. After cleaning, the instrument shall be boiled in distilled water in a glass beaker or other suitable container, for 30 minutes and allowed to remain submerged for an additional 24 hours. There shall be no signs of corrosion on any surface required to be smooth. Corrosion shall not exceed 2% of total maximum surface area on other parts of the instrument. Slight corrosion in serrations, teeth, crevices, etc. shall not be cause for rejection.

4.5.5 Composition analysis. Specimens of materials used to fabricate the instrument or taken from the sample shall be subjected to analysis with methods 111.2 or 112.2 of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning, drying and preservative. Each retractor shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116. Preservative shall not be applied.



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5.1.1.2 Unit package. Each retractor shall be wrapped in neutral tissue or other suitable anticorrosive material. Each retractor shall be further packaged in accordance with MIL-P-116, method IC-2.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Eighteen (18) retractors shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be in accordance with the appendix of the applicable box specification. Case liner shall not be required.

5.2.2 Level B. Eighteen (18) retractors shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities.

5.3.2.1 Unit package and exterior container. Each unit package and exterior container shall be marked in accordance with MIL-STD-129.



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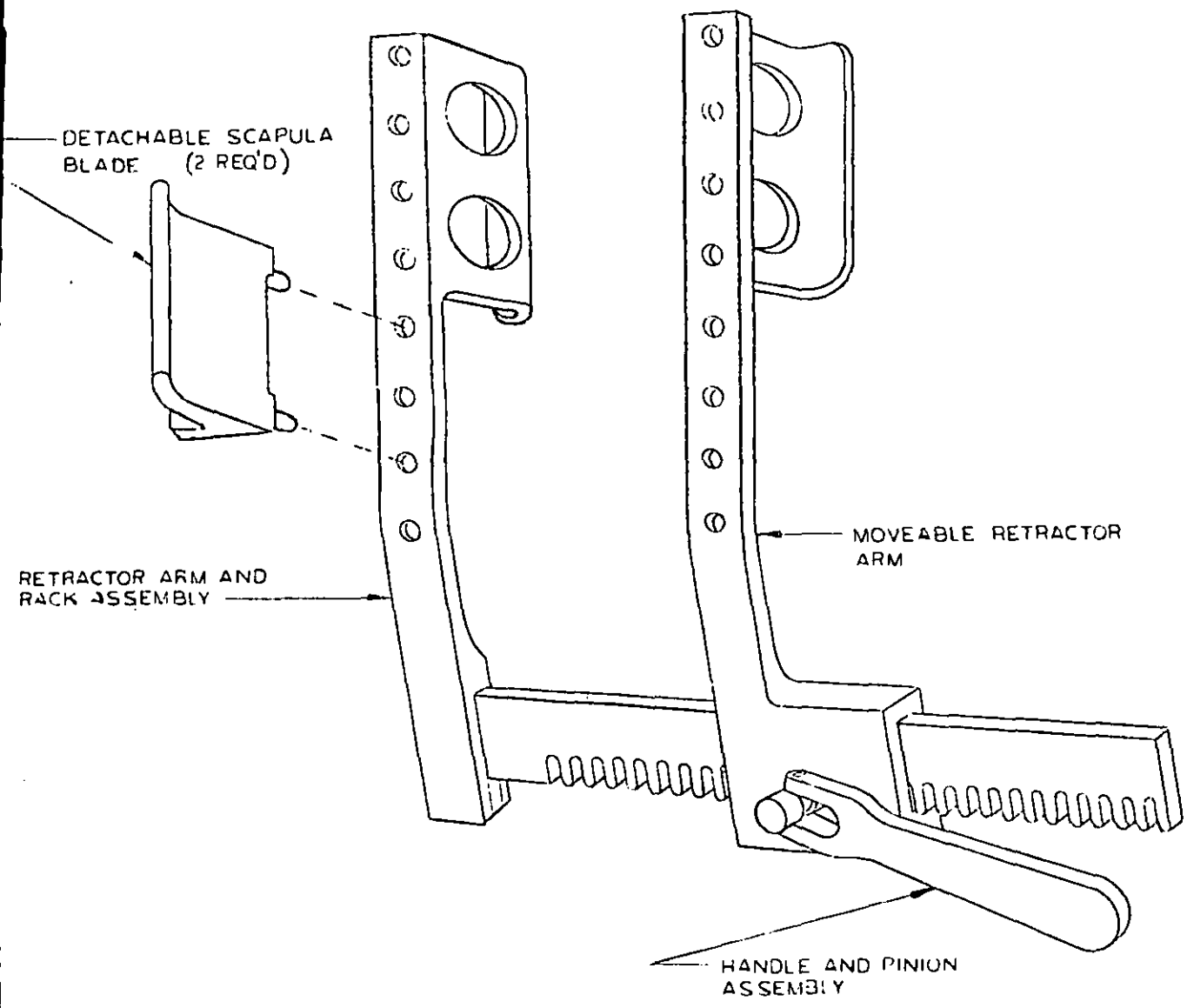



FIGURE 1. Harken, rib retractor assembly



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FIGURE 2. Handle and pinion assembly



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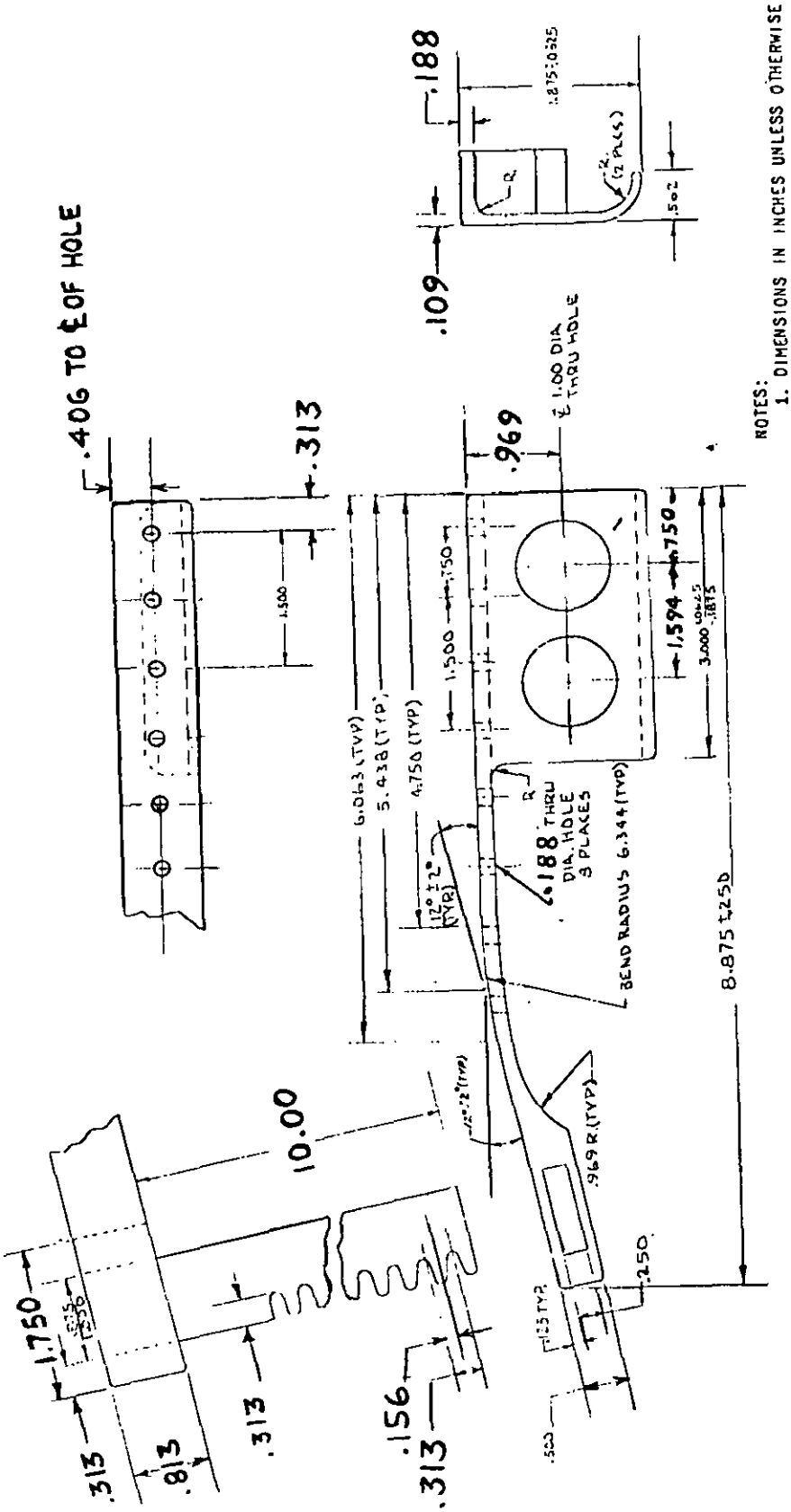
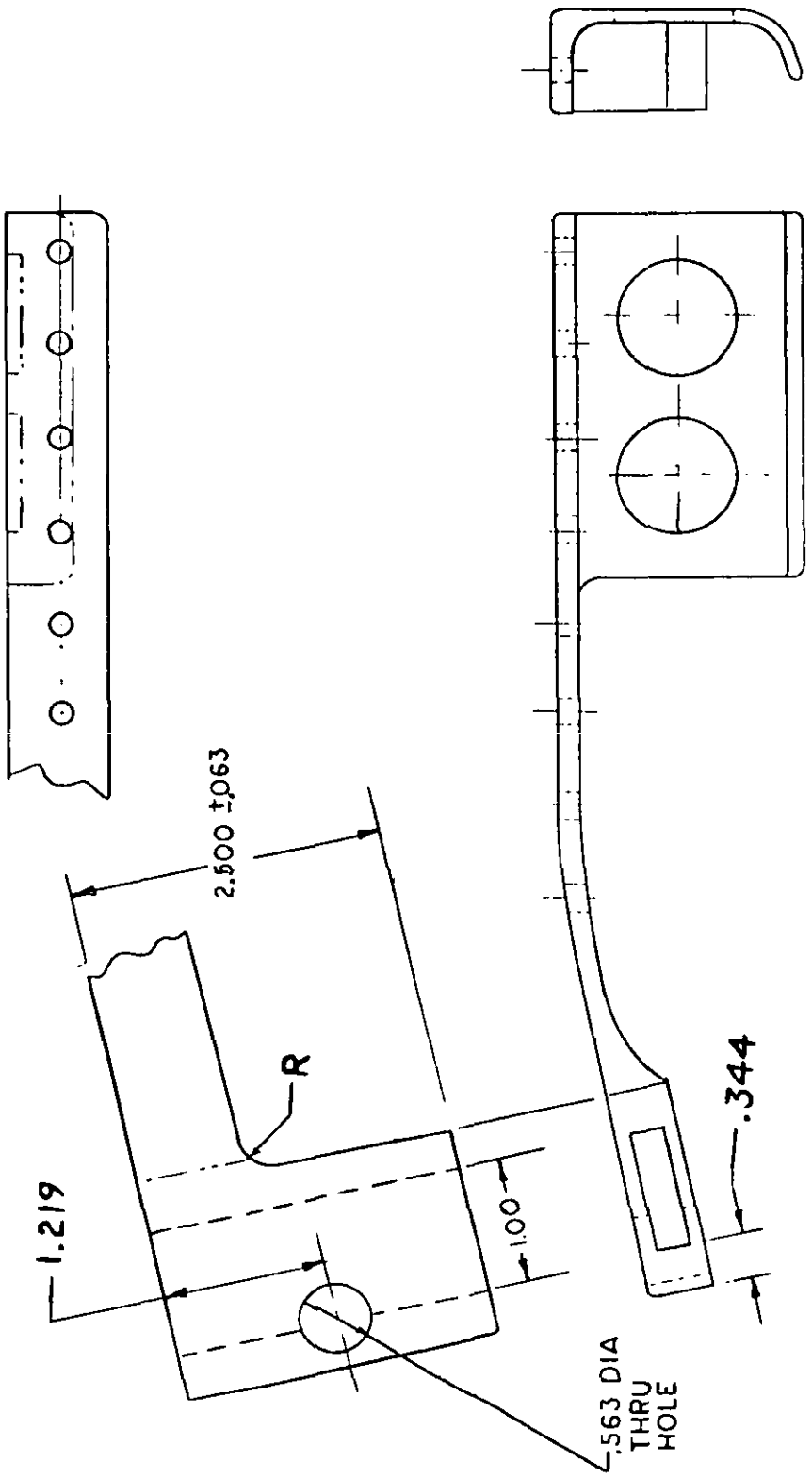



FIGURE 3. Retractor arm and rack

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NOTES:
1. DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED TOLERANCES.
FRACTIONS ± 1/64
DECIMALS ± .015
2. BREAK ALL SHARP EDGES.
3. ALL DIMENSIONS NOT DELINEATED ARE SAME AS FIGURE 3.

FIGURE 4. Moveable arm



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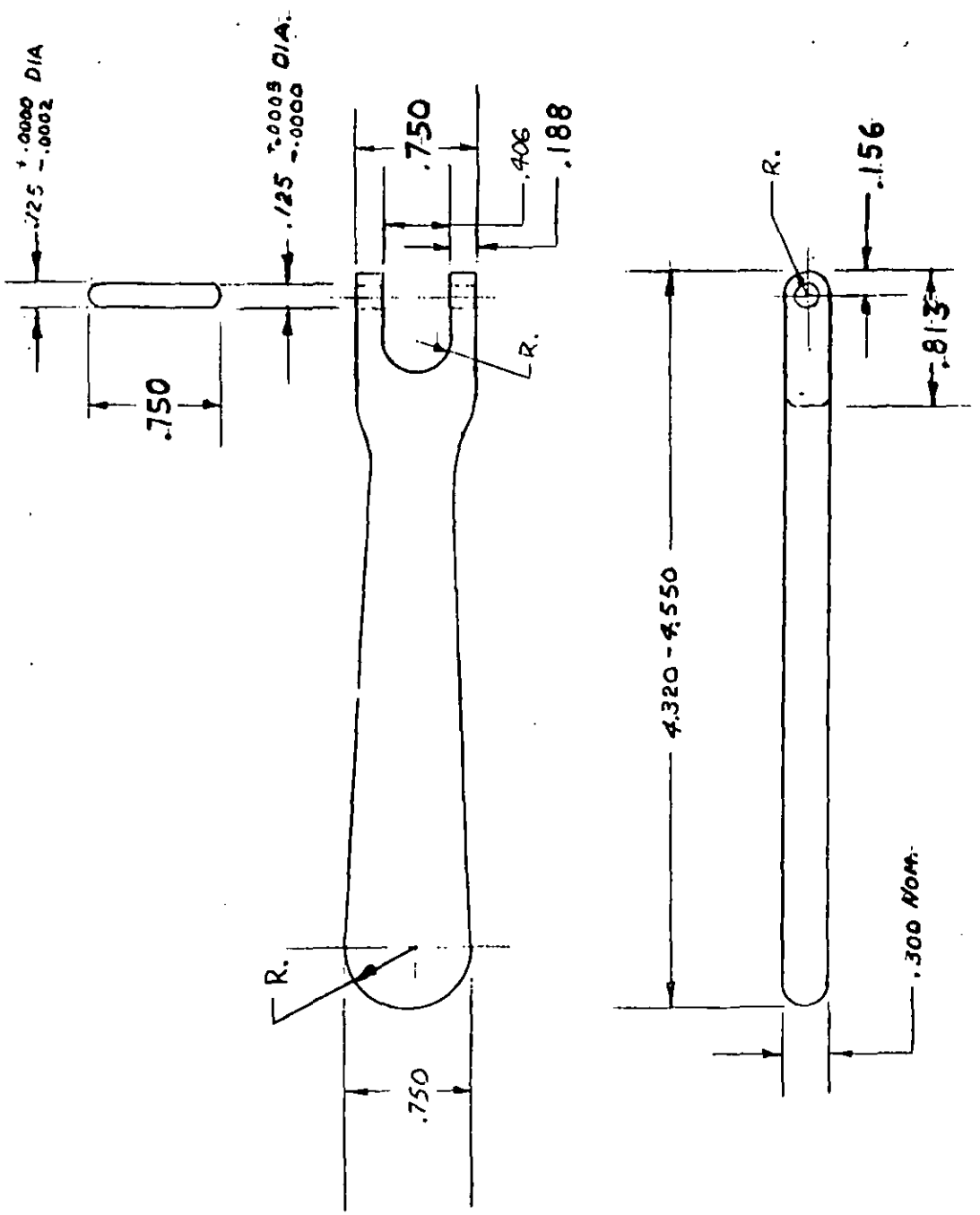
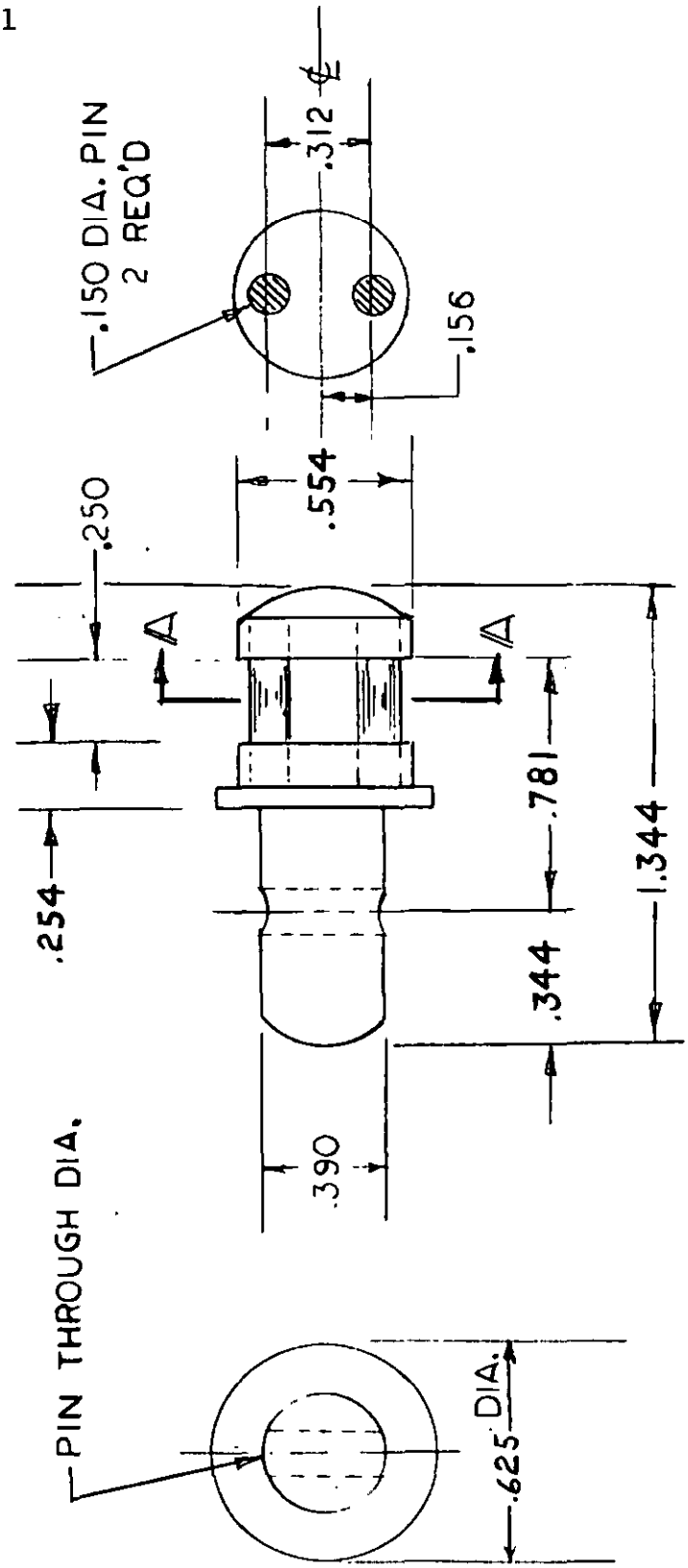



FIGURE 5. Handle and pin

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NOTES:
1. DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED TOLERANCES.
FRACTIONS ± 1/64
DECIMALS ± .015
2. BREAK ALL SHARP EDGES.

FIGURE 6. Pinion



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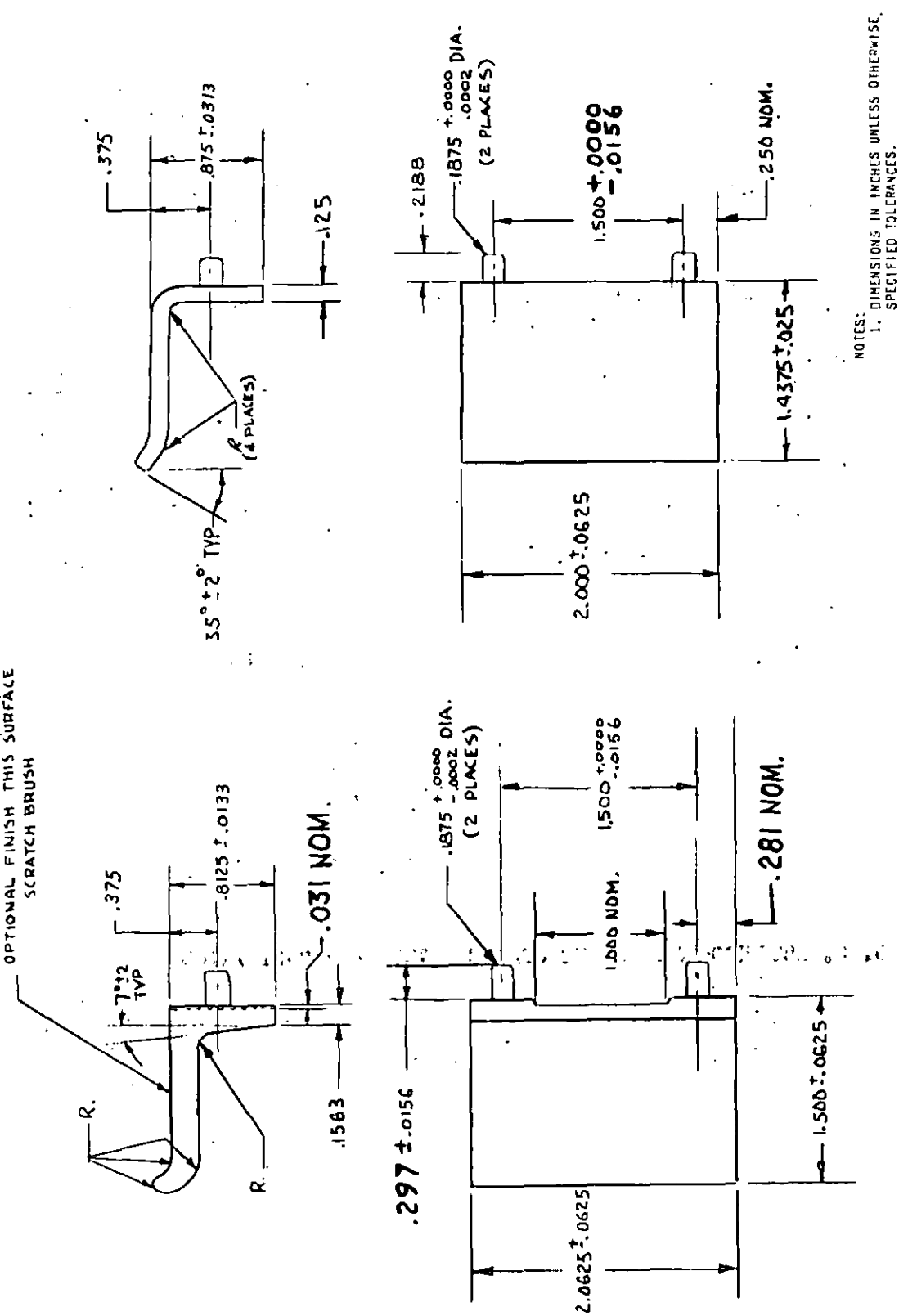



FIGURE 7. Detachable scapula blade



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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

| National Stock No. (NSN) | Item Identification |
|---|---|
| 6515-00-299-8756 | RETRACTOR, RIB, HARKEN |

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

MILITARY CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Civil Agency Coordination Activity
GSA-FSS
VA-DMS
COMMERCE-NBS

Project No. 6515-1954

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 50 cents each.



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
GG-R-1971 RETRACTOR, RIB, HARKEN

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

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